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Systems and methods for automation of end-to-end network slice deployment

Abstract

Systems and methods described herein enable automation of network slice deployments. A network device stores a group of preconfigured templates for different slice segments of a data transport network and identifies customer parameters for a slice. The network device retrieves a first template that satisfies the customer parameters for a first slice segment and generates a first segment design for the slice. The network device retrieves a second template that supports the first segment design and customer parameters for a second slice segment and generates at least one second segment design for the slice. The network device assembles an end-to-end slice design using the first segment design and the at least one second segment design and forwards the end-to-end slice design to a deployment system for the data transport network.

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Background/Summary

BACKGROUND

(1) Next Generation mobile networks, such as Fifth Generation (5G) mobile networks, are being deployed as the next evolution of mobile wireless networks. 5G mobile networks are designed to increase data transfer rates, increase spectral efficiency, improve coverage, improve capacity, and reduce latency. For example, a 5G network may incorporate “network slicing” technology to increase network efficiency and performance.

(2) Network slicing is a type of virtualized networking architecture that involves partitioning of a single physical network into multiple virtual networks. The partitions, or “slices,” of the virtualized network may be customized to meet the specific needs of applications, services, devices, customers, or operators. Each network slice can have its own architecture, provisioning management, and security that supports a particular application or service. Bandwidth, capacity, and connectivity functions are allocated within each network slice to meet the requirements of the particular network slice. Network slicing may be implemented in a dynamic fashion, such that the slices of the virtualized network may change over time and may be re-customized to meet new or changing needs of applications, services, devices, customers, or operators. However, designing and building network slices is currently a labor-intensive process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrate concepts described herein;

(2) FIG. 2 illustrates an example network environment in which end-to-end network slicing deployments may be implemented;

(3) FIG. 3 is a diagram illustrating example components and communications associated with a slice planning tool, according to an implementation;

- (4) FIG. 4 is a diagram illustrating radio access network (RAN) parameters for the slice planning tool, according to an implementation;
- (5) FIG. 5 is a diagram illustrating midhaul parameters for the slice planning tool, according to an implementation;
- (6) FIG. 6 is a flow diagram illustrating an exemplary process for automated end-to-end network slice deployment, according to an implementation; and
- (7) FIG. 7 is a diagram illustrating example components of a device that may correspond to one or more of the devices illustrated and described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (8) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements. Also, the following detailed description does not limit the invention.
- (9) Virtualized radio access networks (RANs) and virtualized transport domains involve the use of network functions virtualization (NFV) and software defined networks (SDNs) to virtualize a portion of the RANs and transport domains on standard information technology (IT) and commercial off-the-shelf (COTS) hardware. Virtualized RANs and transport domains offer a number of advantages, including a flexible and scalable architecture that enables dynamic load-balancing, intelligent traffic steering, and latency reduction using local caching. Virtualization of RANs and transport domains is particularly useful for implementing network slicing. Network slicing, through the use of NFV and SDNs, enables Next Generation mobile networks to offer a variety of services that demand different network performances for different combinations of RAN and transport domains.
- (10) Systems and methods described herein relate to end-to-end design of network slices. The systems and methods enable automation of network slice deployments. Modular parts of basic network infrastructure may be modified and assembled to meet a customer's network slice requirements. A slice planning tool (e.g., operating on a network device or computing device) may store a group of preconfigured templates for different modular parts of a network slice (referred to herein as slice segments) in a data transport network and may identify customer parameters for a new slice. Customer input may govern selection of the preconfigured templates for the different slice segments and the selected templates may be modified to meet customer requirements and conform to configurations of another segment in the customer's slice. Once each slice segment is configured, the slice planning tool may assemble the individual segment designs into an end-to-end slice design for deployment.
- (11) In contrast with implementations described herein, conventional cloud platform planning tools generate an output for a new network slice that a design team enters into a cut sheet and/or spreadsheet file. Data from the cut sheet must be entered (e.g., typically by hand) into an application design tool that generates a remedy ticket for a human to build a tenant space, and then another ticket is run to create new ports for the new network slice. Implementations described herein collect customer requirements and slice definitions once via a slice planning tool that can assemble successive stages and initiate an orchestration process.
- (12) FIG. 1 illustrates network service slice arrangements that may be used with exemplary embodiments described herein. As shown, a network service slice infrastructure design may be implemented across multiple networks, from a core network **100-1**, to an edge network **100-2**, to a far edge network **100-3**, and out to a cell site **100-4** in a Radio Access Network (RAN). The service slice infrastructure design includes different transport domains, such as, for example, backhaul, midhaul, and fronthaul transport domains, which may span the different networks shown in FIG. 1.
- (13) In one example of a network service slice infrastructure design, shown at **110-1**, an application ("App") resides in the core network **100-1** and connects to a User Plane Function ("UPF") in the edge network **100-2** via an application ("APP") transport domain. The UPF in design **110-1** may be co-located, in edge network **100-2**, with a Central Unit-User Plane (CU-UP) (e.g., at a same group

of network devices). The UPF and CU-UP in design **110-1** connect with one another via a backhaul (“BH”) transport domain. The BH transport domain includes the intermediate link(s) between transport nodes (e.g., UPF) in, or close to, the core network and transport nodes (e.g., CU-UP) in one or more edge networks. The CU-UP in edge network **100-2** connects to a RAN Distributed Unit (DU) via a midhaul (“MH”) transport domain located in far edge network **100-3**. The MH transport domain includes the intermediate link(s) between transport nodes (e.g., CU-UPs) close to the core network and the radio controllers (e.g., DUs) in a wireless network. The DU connects to a RAN Remote Unit (RU) at the RAN cell site **100-4** via a fronthaul (“FH”) transport domain. The FH transport domain includes the intermediate link(s) between the radio controllers (e.g., DUs) and the radio heads (e.g., RUs) in a wireless network.

(14) In another example of a network service slice infrastructure design, shown at **110-2**, an application resides in the core network **100-1** and connects to a UPF in the edge network **100-2**. The UPF in design **110-1** may be co-located, in edge network **100-2**, with a CU-UP (e.g., at a same group of network devices **215** shown in FIG. 2). The UPF and CU-UP in design **110-2** connect with one another via a BH transport domain. The CU-UP in edge network **100-2** connects to a DU, located at a RAN cell site **100-4**, via a MH transport domain. The DU connects to a RU at the RAN cell site **100-4** via a FH transport domain. The DU and RU in design **110-2** may be co-located, at RAN cell site **100-4**.

(15) Other designs, having different configurations than those shown, may be used with the exemplary embodiments described herein. For example, in other arrangements, an application may be located with the UPF and CU-UP in edge network **100-2** to provide low-latency communications (LLC) or in far edge network **100-3** to provide ultra-low-latency communications (ULLC).

(16) In the example of FIG. 1, RAN cell site **100-4** may be included in a 5G RAN, a future generation RAN (e.g., a Sixth Generation (6G) RAN, a Seventh Generation (7G) RAN, or a subsequent generation RAN), a centralized-RAN (C-RAN), an Open Radio Access Network (O-RAN), and/or another type of radio access network. The RAN may be implemented according to service levels and various wireless standards, frequencies, bands, and segments of radio spectrum, and/or other attributes or technologies used for radio communication. For example, a customer may have different service level options that are primarily controlled at the RAN level and can be provided through different network slices. Slice subscriptions for a customer may include, for example, different purchase options and performance criteria (e.g., latency, throughput, reliability, availability, redundancy, etc.) available for a requested network slice.

(17) Next Generation mobile networks typically include multiple transport domains, such as, for example, FH, MH and BH transport domains, with each transport domain having its own unique and complex characteristics. Each transport domain may operate to serve one or more network slices. Network infrastructure design may include multiple different transport domains that interconnect and operate to service the traffic of one or more network slices. While deploying and managing network slices for customers with specific service requirements (e.g., specific Service Level Agreements (SLAs)), determining which transport domain to orchestrate, manage, and control to optimally and most efficiently maintain those requirements becomes a complicated proposition. Other aspects of network slicing that may need to be defined for each customer's network slice may include different charging profiles, roaming options, reporting standards, security levels, exposure requirements, operations support system (OSS) or business support system (BSS) parameters, and the like.

(18) FIG. 2 illustrates an exemplary network environment **200** in which end-to-end network slicing deployment, as described herein, may be implemented. Network environment **200** includes a data transport network section **205** and a network management network section **210**. Data transport network section **205** may include multiple groups of network devices (or data centers) **215-1** through **215-5**, interconnected via respective transport domains **220-1** through **220-4**. In the

exemplary implementation shown in FIG. 2, an application **225** may be installed at network device **215-1**, a UPF **230** may be installed at network device **215-2**, a CU-UP **235** may be installed at network device **215-3**, a DU **240** may be installed at network device **215-4**, and a RU **245** may be installed at network device **215-5**.

(19) The application **225** may include any type of application that sends data to, and/or receives data from, a user equipment device (UE), such as UE **201**, wirelessly connected to a RAN cell site. Application **225** may be installed at data center **215-1**, or in another network device connected to core transport domain **220-1**. UPF **230** includes a logical node that hosts packet routing and forwarding, policy enforcement, and data buffering functions, and includes functionality for interconnecting with an external Data Network (not shown) (e.g., the Internet). In one implementation, UPF **230** may be implemented as a Virtual Network Function (VNF) or Cloud-native Network Function (CNF) and installed at data center **215-2**. CU-UP **235** includes a logical node that hosts user plane functions, such as, for example, data routing and transport functions. In one implementation, CU-UP **235** may be implemented as a VNF or CNF and installed at data center **215-3**.

(20) DU **240** includes a logical node that hosts functions associated with the Radio Link Control (RLC) layer, the Medium Access Control (MAC) layer, and the physical (PHY) layer. In one implementation, DU **240** may be implemented as a VNF or CNF and installed at data center **215-4**. RU **245** may include at least one radio transceiver, and associated antenna(s), for radio frequency (RF) wireless communication with one or more UEs **201**. RU **245** may additionally include a logical node that operates and controls the at least one radio transceiver. In one implementation, the logical node of RU **245** may be implemented as a VNF or CNF and installed at data center **215-5**. DU **240** and RU **245** together may be components of a New Radio (NR) Next Generation gNodeB used in the RAN of a Next Generation mobile network, such as, for example, a 5G mobile network. UE **201** may include any type of electronic device having a wireless capability (e.g., a Radio Frequency (RF) transceiver) to communicate with RU **245**. Each of UEs **201** may include, for example, a computer (e.g., desktop, laptop, tablet, or wearable computer), a personal digital assistant (PDA), a “smart” phone, or a “Machine-to-Machine” (M2M) or “Internet of Things” (IoT) device. A “user” (not shown) may own, operate, and/or administer each UE.

(21) As shown in FIG. 2, core transport domain **220-1** may interconnect application **225** and UPF **230**, backhaul transport domain **220-2** may interconnect UPF **230** and CU-UP **235**, midhaul transport domain **220-3** may interconnect CU-UP **235** and DU **240**, and fronthaul transport domain **220-4** may interconnect DU **240** and RU **245**. Core transport domain **220-1**, backhaul transport domain **220-2**, midhaul transport domain **220-3**, and fronthaul transport domain **220-4** may each include a transport network that further includes one or more links for interconnecting respective network devices **215**. In the network environment **200** of FIG. 2, data may be transported, for example, from application **225** to RU **245** (for wireless transmission to a destination UE **201**) via UPF **230**, CU-UP **235**, DU **240** and via the interconnected transport domains **220-1** through **220-4**. Furthermore, data (e.g., received from a UE **201** via wireless transmission) may be transported from RU **245** to application **225** via DU **240**, CU-UP **235**, UPF **230** and via the interconnected transport domains **220-4** through **220-1**.

(22) Network management section **210** of network environment **200** may include a customer ordering platform **250**, a slice planning tool **260**, a parameter forms database (DB) **270**, a design system interface **280**, and a deployment system **290**.

(23) Customer ordering platform **250** may include network devices that provide a web-based interface for a customer (e.g., using a user device) to access a slice ordering service. Via the user device, users (e.g., customers) of data transport network section **205** may access customer ordering platform **250** to manage (e.g., order, configure, issue commands, update, monitor, etc.) network slices for new applications (e.g., applications **225**) or services, for example. Using customer ordering platform **250**, customers may enter their network slice/service configurations with selected

eligible parameters supported by data transport network section **205**. According to an implementation, customer ordering platform **250** may use a configuration information questionnaire (CIQ) as an initial inquiry tool. The CIQ typically lists possible NF or service attributes, which a vendor can select for setting on a per-site basis or a per-feature basis.

(24) Slice planning tool **260** may include an automated platform to design slice components, including network function designs, service designs, and slice designs. A network function design may include a VNF or CNF that may be used to support a service. A service design may be an assembly of multiple network functions to support a given application or set of requirements for a network service. A slice design may be an end-to-end network path of network components and/or services. A resultant product from slice planning tool **260** may be a final slice service package that contains pieces, such as service, network functions, and associated slice configurations applicable to each network function that makes up a slice. In one implementation, the final slice service package may be a cloud services archive (CSAR) package (e.g., a ZIP file). The configuration design may be performed on the basis of a slice functioning in a particular environment (e.g., a site-specific configuration), such a particular data center for a network function virtualization infrastructure (NFVI) and/or a cloud computing platform.

(25) According to one implementation, slice planning tool **260** may provide slice/service options for presentation on customer ordering platform **250** and receive customer selections. As described further herein, slice planning tool **260** may provide standardized network options and work through an iterative process to define a customer order. For example, slice planning tool **260** may assemble customized end-to-end slice designs from templates of different network components and transport domains (i.e., network slice segments) based on customer inputs.

(26) Parameter forms DB **270** may include form templates associated with network slice services. Parameter forms DB **270** may include memory to store a group of preconfigured templates for different network slice segments of a data transport network. For example, parameter forms DB **270** may include tables or lists of different service requirements and/or specifications that can be entered by a customer to define a slice order. Individual parameter forms/templates may be defined, for example, for different network slice segments (e.g., RAN, midhaul transport, service aggregation point (SAP), transport aggregation point (TAP), etc.) as described further herein. Slice planning tool **260** may retrieve appropriate parameter forms from parameter forms DB **270** in response to customer input during an ordering process.

(27) Design system interface **280** may provide communications to different network discover tools and/or design systems. For example, design system interface **280** may include one or more application programming interfaces (APIs) to interface with a network management system for network mapping and device discovery. According to one implementation, slice planning tool **260** may use design system interface **280** to query a network physical infrastructure, performance, and assigned usage. According to another implementation, slice planning tool **260** may use design system interface **280** to supplement information from parameter forms DB **270** (e.g., in response to customer input). For example, design system interface **280** may initiate an updated network map or verification of performance criteria to support a customer slice in a particular geographic area.

(28) Deployment system **290** may include one or more network devices or computer devices to design and orchestrate network slices in data transport network section **205**. Deployment system **290** may receive instructions from slice planning tool **260** that identify one or more transport domains **220** and/or VNFs/CNFs to orchestrate within data transport network section **205**.

Deployment system **290**, based on the instructions from slice planning tool **260**, may initiate a process of orchestrating particular transport domains and/or VNFs/CNFs in data transport network section **205**.

(29) Deployment system **290** may include multiple sub-components such as, for example, a slice inventory, a runtime slice orchestrator, a network function virtualization orchestrator (NFVO), and a virtual network function manager (VNFM). The slice inventory may include memory or database

with an inventory of active (e.g., instantiated) slices associated with data transport network section **205**. The runtime slice orchestrator may manage creation of network slices using information from slice planning tool **260**. For example, the runtime slice orchestrator may be responsible for implementing instructions to instantiate, allocate, and activate network slices. The NFVO may create, remove, and/or update network functions, such as virtualized versions of core components (e.g., session management function (SMF), UPF, etc.) at a particular site (e.g., a data center, enterprise, multi-access edge computing (MEC), etc.). The VNFM may perform lifecycle management of particular VNFs, including instantiation, scaling, updating, and/or termination of particular VNFs based on instructions received from, for example, the runtime slice orchestrator.

(30) The configuration of network components of network environment **200** shown in FIG. 2 is for illustrative purposes. Other configurations may be implemented. Therefore, network environment **200** may include additional, fewer and/or different components, that may be configured in a different arrangement, than that depicted in FIG. 2. For example, though application **225**, UPF **230**, CU-UP **235**, DU **240**, and RU **245** are shown as being installed at different network devices **215**, one or more of application **225**, UPF **230**, CU-UP **235**, DU **240**, and RU **245** may be installed at a same network device **215**. Additionally, though backhaul transport domain **220-2**, midhaul transport domain **220-3**, and fronthaul transport domain **220-4** are shown in FIG. 2 as being separate networks, that may have multiple links, in some implementations, these domains may be implemented as buses within a single network device **215**, or as a short distance link (e.g., a short run of optical fiber) between two network devices **215**.

(31) FIG. 3 is a diagram illustrating example components and communications associated with slice planning tool **260**. As shown in FIG. 3, slice planning tool **260** may obtain iterative input **302**, such as user input solicited and obtained via customer ordering platform **250**. Slice planning tool **260** may obtain from parameter forms DB **270** rules and/or templates **304** to solicit different network slice parameters from customers and to assemble network slice segments. Slice planning tool **260** may also obtain, via design system interface **280**, various network data **306** to supplement and/or confirm information in parameter forms DB **270** and compatibility with customer input **302**.

(32) As shown in FIG. 3, parameters may be defined for different network segments and features, such as RAN parameters **305**, midhaul parameters **310**, SAP transport rules **315**, charging parameters **320**, application transport rules **325**, roaming rules **330**, reporting rules **335**, wholesale rules **340**, security and services **345**, exposure function **350**, and OSS/BSS rules **355**. Regarding RAN parameters **305**, slices may be configured with different types of latency, throughput, reliability, availability, redundancy, etc. Some of these performance criteria may be grouped by a Quality of Service (QoS) level, a QoS Class Identifier (QCI for Non-Standalone Architecture)/5G QoS Identifier (5QI for Standalone Architecture), and/or different scheduling priorities. These QoS, QCI/5QI, and scheduler systems may be defined, and set to, for example, geographic areas as another design parameter. Customers, using customer ordering platform **250**, may input and/or select RAN parameters to meet desired specifications for network slices to support, for example, a new application.

(33) RAN parameters **305** may include, for example, parameters illustrated in FIG. 4. As shown in FIG. 4, RAN parameters **305** may include a scheduler field **405**, a Quality of Service (QoS) field **410**, a QoS Class Identifier (QCI, for Non Stand Alone Architecture) field **415**, a 5G QoS Identifier (5QI, for Stand Alone Architecture) field **420**, a geography field **425**, a cell type field **430**, a spectrum type field **435**, a resource type field **440**, a default priority field **445**, a packet delay budget field **450**, a packet error rate field **455**, a default maximum data burst volume field **460**, a default averaging window field **465**, a notification control field **470**, and a Differentiated Services Code Point (DSCP) field **475**.

(34) Scheduler field **405** may indicate logic that schedules and assigns the required resource to network slices of access network **105**. For example, a RAN scheduler may schedule and assign network resources of a RAN. As an example, in the RAN, radio resources may be allocated in time

with time-division multiplexing (TDM), or in frequency with frequency-division multiplexing (FDM), or in space with space-division multiplexing (SDM), or some combination thereof, using centralized or decentralized dynamic scheduling across network slices. A scheduler may schedule and assign resources for RAN slices according to SLA requirements, QoS requirements, metric requirements, and/or other criterion pertaining to the access network, which may or may not be different from other networks (e.g., core network, etc.).

(35) QoS field **410** may include a quality of service information for a requested slice. QCI field **415** and/or 5QI field **420** may indicate numerical QoS level corresponding to QoS field **410**. Geography field **425** may include geographic service location information for requested slice access (e.g., local, regional, etc.). Geographic service location information may include, for example, tracking area IDs (TAIs), E-UTRAN cell global identifiers (ECGIs), enterprise network IDs, or other location parameters.

(36) Cell site type field **430** may indicate a cell type (e.g., macro-cell, micro-cell, etc.). Spectrum type field **435** may indicate various frequencies, bands, and segments of radio spectrum (e.g., centimeter (cm) wave, millimeter (mm) wave, below 6 gigahertz (GHz), above 6 GHz, higher than mm wave, etc.). Spectrum type field **435** may include a list of available frequency bands for the current location specified in geography field **425**.

(37) Resource type field **440** may include a resource type to be used by the network slice in the RAN (e.g., delay-critical guaranteed bit rate (GBR), non-GBR, etc.). Default priority level field **445** may include a default priority value for network slice traffic. Packet delay budget field **450** may include a packet delay budget value for the slice. Packet error rate field **455** may include a packet error rate value. Default Maximum Data Burst Volume field **460** may include a default maximum data burst volume value. Default Averaging Window field **465** may include a default averaging window value associated with the network slice performance. Notification Control field **470** may include whether a QoS notification control indication (e.g., a QNC indication) may be used to instruct a base station to monitor whether measured QoS parameters meet the slice requirements. DSCP field **475** may include a DSCP value used for prioritizing traffic within a network.

(38) Referring back to FIG. 3, slice planning tool **260** may obtain initial user input **302** to identify required parameters for a network slice. The customer required parameters may define basic service requirements for network services, such as a geographic service location (e.g., a region, an enterprise network location, a data center, etc.) and a performance criteria (e.g., data speeds, quality of service (QoS) levels, required bandwidth, latency, priority, etc.). In response to the customer required parameters, slice planning tool **260** may obtain, via communication **304**, a RAN template from parameter forms DB **270** that meets the customer required parameters. For example, RAN designers (e.g., network engineers) may design RAN templates for different slice types which may be populated into parameter forms DB **270**. Each RAN template may define, for example, where schedulers manage the RF in certain bands, on a certain footprint, with upload speeds and download speeds that are supported.

(39) After selecting a RAN template, slice planning tool **260** may request, if needed, additional user input **302** to complete the required RAN parameters **305** for the RAN template. With the completed RAN parameters **305**, slice planning tool **260** may proceed to obtain midhaul parameters **310**.

(40) Midhaul parameters **310** may include parameters for midhaul transport **220-3**. Midhaul transport designers (e.g., network engineers) may design transport rules and parameters for different slice types which may be populated into parameter forms DB **270**. Midhaul transport **220-3** may include a link between a DU (e.g., DU **240**) and the CU (e.g., CU-UP **235**) via, for example, an F1 interface. According to one implementation, the DU may be at the cell or hub and the CU may be in the SAP. Regarding mid-haul parameters **310**, there may be several parameters that can be modified in the midhaul transport. One such parameter may be the CU end points, another may be the DSCP bits that define how traffic is treated and the type of traffic, such as non-GBR and

GBR. Other factors may be the types of transport that are available in a selected region—such as fiber, Ethernet, or microwave—and the slice ID and single network slice selection assistance information (S-NSSAI) value. Slice planning tool **260** may request, if needed, additional user input **302** to complete the required midhaul parameters **310**.

(41) Midhaul parameters **310** may include, for example, parameters illustrated in FIG. 5. As shown in FIG. 5, midhaul parameters **310** may include a connectivity type field **505**, a QoS field **510**, a QCI field **515**, a 5QI field **520**, a geography field **525**, DSCP field **530**, a tagging field **535**, a resource type field **540**, a default priority level field **545**, a packet delay budget field **550**, a packet error rate field **555**, a default maximum data burst volume field **560**, a default averaging window field **565**, a slice ID field **570**, a S-NSSAI field **575**, a GBR field **580**, a non-GBR field **585**, a queuing field **590**, a link aggregation group (LAG) field **595**, and a non-LAG field **599**.

(42) Connectivity type field **505** may indicate, for example, a type of connectivity for a midhaul link, such as fiber, Ethernet, or microwave, direct connectivity, or indirect connectivity, etc. QoS field **510**, QCI field **515**, 5QI field **520**, geography field **525**, and DSCP field **530** may correspond to fields **410**, **415**, **420**, **425**, and **475** of RAN parameters **305**. Tagging field **535** may indicate any tagging requirements for protocol data units (PDU) using the network slice. Resource type field **540** may include a resource type to be used by the network slice in the midhaul transport (e.g., delay-critical GBR, non-GBR, standard GBR, etc.).

(43) Default priority level field **545** may indicate a default priority level (e.g., a ranked level between 1 and 8) for the network slice. Packet delay budget field **550**, packet error rate field **555**, default maximum data burst volume field **560**, and default averaging window field **565** may each include values relating to performance metrics for the network slice. According to one implementation, values for packet delay budget field **550**, packet error rate field **555**, default maximum data burst volume field **560**, and default averaging window field **565** may correspond to certain values of QCI field **515** and/or 5QI field **520**.

(44) Slice ID field **570** may store a unique identifier for the network slice. In one implementation, the unique ID stored in Slice ID field **570** may correspond to an IP address for an access and mobility management function (AMF) instance that supports the network slice. S-NSSAI field **575** may store data that includes S-NSSAI. For example, S-NSSAI field **575** may include data that indicates a slice/service type (SST) value. The data may also include slice differentiator (SD) information, for example. GBR field **580** and non-GBR field **585** may store a GBR value or non-GBR value associated with the particular QoS class. Queuing field **590** may define a queue type/size. LAG field **595** and non-LAG field **599** may indicate whether the network slice can be associated with a link aggregation group.

(45) Slice planning tool **260** may perform network discovery to map a network environment. Slice planning tool may map the RAN QCI/5QI's (e.g., from QCI field **415** and/or 5QI field **420**) to DSCP parameters from the Cell Site router to the TAP. As an initial step, slice planning tool **260** may map each cell site to each transport aggregation point. According to an implementation, slice planning tool **260** may also identify any intermediate aggregations points. In one implementation, slice planning tool **260** may use Common Language Location Identification (CLLI) codes to stitch together the constant bit rate (CBR) to the TAP sites. Next slice planning tool **260** may map the CBR from the TAP sites to the SAP sites. Slice planning tool **260** may map DSCP to Port Control Protocol (PCP) parameters.

(46) Once a mapping of the network environment is available, slice planning tool **260** may apply midhaul transport rules from parameter forms DB **270**. In one implementation, parameter forms DB **270** may include YANG files to define the structure and NETCONF files to configure the elements in the map.

(47) SAP transport rules **315** provide rules for the link between the midhaul transport and the cloud platform infrastructure, along with the inter element communications. SAP transport rules **315** may include rules on how network elements (e.g., SMF, UPF, AMF, etc.) of the network slice are

interconnected. As an example, SAP transport rules **315** may define the connection from a SMF to a UPF, firewall rules for a new SMF/UPF pair. SAP transport rules **315** may also identify locations from where to pull IP addresses, which may be used to instantiate the virtual machines for each network function. SAP transport rules **315** may include information to program segment routing (SR) fabric, such as routing to a firewall to Internet and back.

(48) Charging parameters **320** may include one or more charging plan templates for customers to apply to a network slice. Examples of different charging plans may include sponsored data plans, monthly usage plans, charging per time on the network, etc. The different charging templates may be stored, for example, in parameter forms DB **270**. Slice planning tool **260** may obtain user input **302** to identify charging parameters for the network slice and match the parameters to an available charging plan that can be retrieved as a template **304** from parameter forms DB **270**.

(49) Application transport rules **325** may identify a type of data traffic that the network slice is required to support. Examples of different traffic types include IPv6, IPv4, dual stack, Network Address Translated (NAT), Non-NAT, Non-IP Data Delivery, etc. Different application transport templates for the different traffic types may be stored, for example, in parameter forms DB **270**. Slice planning tool **260** may obtain user input **302** to identify application transport parameters for the network slice and match the parameters to an available transport rules template that can be retrieved as a template **304** from parameter forms DB **270**.

(50) Roaming rules **330** may identify roaming profiles for the network slice. For example, roaming profiles may include permissions to allow or not allow roaming on 4G or 5G networks while using the network slice, or restrictions for roaming. Different roaming profile templates may be stored, for example, in parameter forms DB **270**. Slice planning tool **260** may obtain user input **302** to identify application transport parameters for the network slice and match the parameters to an available roaming template that can be retrieved as a template **304** from parameter forms DB **270**.

(51) Reporting rules **335** may identify customer requirements for reporting. Reporting rules **335** may define report scope (e.g., slice level, enterprise level, etc.), frequency (e.g., weekly, monthly, etc.), content (e.g., QoS, usage statistics, etc.), and the like. Different reporting templates may be stored, for example, in parameter forms DB **270**. Slice planning tool **260** may obtain user input **302** to identify reporting parameters for the network slice and match the parameters to an available reporting template that can be retrieved as a template **304** from parameter forms DB **270**.

(52) Wholesale rules **340** may provide rules governing “wholesale” arrangements, where one telecommunications carrier may purchase services from another telecommunications carrier and resell such services to retail customers. Different wholesale profiles may be stored, for example, in parameter forms DB **270**. Slice planning tool **260** may obtain user input **302** to identify wholesale parameters for the network slice and match the parameters to an available wholesale profile that can be retrieved as a template **304** from parameter forms DB **270**.

(53) Security and services **345** may include settings for security and privacy-related services, such as firewalls, content filtering, anti-virus scanning, malware protection, etc. Other services may include protected IP pools, traffic shaping (e.g., video optimization, etc.), data speed reductions for usage caps, etc. Templates for different services/combinations may be stored, for example, in parameter forms DB **270**. According to one implementation, features of security and services **345** may be implemented via domain name system (DNS) response police zones. In other implementations, features of security and services **345** may be implemented using an application function. Slice planning tool **260** may obtain user input **302** to identify different security and service requirements for the network slice and match the requirements to available security/service templates that can be retrieved as a template **304** from parameter forms DB **270**.

(54) Exposure function **350** may provide rules governing exposure of network slice data to outside of a provider network. Different exposure profiles may be stored, for example, in parameter forms DB **270**. Slice planning tool **260** may obtain user input **302** to identify exposure parameters for the network slice and match the parameters to an available exposure profile that can be retrieved as a

template **304** from parameter forms DB **270**. According to one implementation, features of exposure function **350** may be implemented via a network exposure function (NEF).

(55) OSS/BSS rules **355** may provide parameters for different OSS/BSS systems for the network slice. An OSS may include a network monitoring system, a network provisioning system, a network management system, a testing system, and/or other types of OSS. A BSS may include an ordering system, a customer service system, a billing system, or the like. Different OSS/BSS profiles may be stored, for example, in parameter forms DB **270**. Slice planning tool **260** may obtain user input **302** to identify different OSS/BSS parameters for the network slice and match the parameters to an available OSS/BSS profile that can be retrieved as a template **304** from parameter forms DB **270**.

(56) Design engine **360** may incorporate customized templates from components **305-355** to generate a slice design for the network slice. The slice design may be an end-to-end network path of network components and/or services that contains pieces, such as service, NFs, and associated slice configurations applicable to each NF that makes up a slice. In one implementation, the final slice service package may be a TOSCA (Topology and Orchestration Specification for Cloud Applications) cloud services archive (CSAR) package **362**. Design engine **360** may forward the CSAR package to deployment system **290** for instantiation.

(57) FIG. **6** is a flow diagram illustrating an exemplary process **600** for automated end-to-end network slicing deployment, according to implementations described herein. In one implementation, process **600** may be implemented by slice planning tool **260**. In another implementation, process **600** may be implemented by slice planning tool **260** in conjunction with one or more other devices in network management section **210**, such as customer ordering platform **250**, parameter forms DB **270**, design system interface **280**, and/or deployment system **290**.

(58) Process **600** may include storing templates and/or rules for standardized network slice segments (block **605**). For example, network teams (e.g., design/engineering teams) may generate a set of standard templates or rule sets for different network segments (e.g., RAN segment, midhaul transport segment, SAP transport segment, application transport segment, fronthaul segment, security and services segments, etc.). As an example, network teams may design RAN templates for different service levels. Each standard RAN template may have a set of parameters that may be supplemented and/or configured as part of a customer ordering process. As another example, network teams may develop a rules set for mapping midhaul transport links to support a RAN segment design. The standard templates and/or rules may be stored, for example, in parameter forms DB **270**.

(59) Process **600** may further include obtaining user parameters for a new network slice (block **610**). For example, a customer may log into customer ordering platform **250** to initiate a new network slice (e.g., to support services for a third party application, enterprise services, etc.). Customer ordering platform **250** may provide a user interface to solicit customer requirements, such as a geographic service location and performance criteria. Customer ordering platform **250** may forward the customer requirements to slice planning tool **260**.

(60) Process **600** may also include matching the user parameters to a first segment template (block **615**) and determining if the parameters for the first network slice segment are complete (block **620**). For example, slice planning tool **260** may match the customer requirements to one of the standard templates for a first segment. Slice planning tool **260** may retrieve, from the group of preconfigured templates in parameter forms DB **270**, a template that satisfies the customer parameters for a first network slice segment. That is, slice planning tool **260** may find a best fit template or a template using the least amount of network resources to satisfy the customer requirements. For subsequent (i.e., next) segments, slice planning tool **260** may find a best fit template to satisfy the customer requirements and support the previous segment designs. In one implementation, when the first slice segment corresponds to a RAN, slice planning tool **260** may determine if the initial customer requirements provide sufficient detail to complete all the necessary

RAN parameters **305** for the selected template. In another implementation, when a second slice segment corresponds to a midhaul transport, slice planning tool **260** may determine if the initial customer requirements provide sufficient detail to complete all necessary midhaul parameters **310** for the selected template and are compatible with the RAN segment design.

(61) If the parameters for the first network slice segment are not complete (block **620**—No), process **600** may include obtaining user input for missing parameters (block **625**). For example, if the initial customer requirements do not provide sufficient detail to complete all the necessary parameters for the selected template, slice planning tool **260** may request additional customer input via customer ordering platform **250**. For example, slice planning tool **260** may request additional RAN parameters **305**. In another implementation, slice planning tool **260** may use design system interface **280** to obtain additional network information needed to obtain additional information for the network slice segment.

(62) If the parameters for the first network slice segment are complete (block **620**—Yes), process **600** may include generating a first network slice segment design (block **630**) and determining if additional network slice segments are needed (block **635**). For example, if the initial customer requirements or supplemented customer requirements provide sufficient detail to complete all the necessary parameters for the selected RAN template, slice planning tool **260** may generate a RAN segment design. Slice planning tool **260** may then determine if additional segment are needed to complete the network slice design for the customer. According to an implementation, slice planning tool **260** may proceed through an ordered sequence to design modular slice components (or segments) for the RAN, midhaul transport, SAP transport, application transport, charging, security, exposure, and OSS/BSS.

(63) If additional network slice segments are needed (block **635**—Yes), process **600** may return to process block **615** to match user parameters to the next network slice segment template. For example, after completing a RAN segment design for the new network slice, slice planning tool **260** may select a rule set or template for midhaul parameters **310**. Furthermore, if a midhaul transport segment design is completed, slice planning tool **260** may select appropriate SAP transport rules **315**, and so forth.

(64) If no additional network slice segments are needed (block **635**—No), process **600** may include assembling the network slice segment designs for deployment (block **640**) and forwarding a slice deployment package to a deployment system for implementation (block **645**). For example, if parameters for all the needed segments have been obtained and the individual segment designs completed, slice planning tool **260** may compile the individual segment designs into a package for deployment (e.g., a CSAR file). Slice planning tool **260** may forward the compiled package to deployment system **290** for instantiation/deployment within data transport network section **205**.

(65) FIG. 7 illustrates example components of a device **700** according to an implementation described herein. Components of network devices **215**, customer ordering platform **250**, slice planning tool **260**, parameter forms DB **270**, design system interface **280**, deployment system **290**, and other network devices in network environment **200** may each include or be implemented on one or more devices **700**. Device **700** may include a bus **710**, a processor **720**, a memory **730**, an input component **740**, an output component **750**, and a communication interface **760**.

(66) Bus **710** may include a path that permits communication among the components of device **700**. Processor **720** may include a processor, a microprocessor, or processing logic that may interpret and execute instructions. Memory **730** may include any type of dynamic storage device that may store information and instructions, for execution by processor **720**, and/or any type of non-volatile storage device that may store information for use by processor **720**. Input component **740** may include a mechanism that permits a user to input information to device **700**, such as a keyboard, a keypad, a button, a switch, etc. Output component **750** may include a mechanism that outputs information to the user, such as a display, a speaker, one or more light emitting diodes (LEDs), etc.

(67) Communication interface **760** may include a transceiver that enables device **700** to communicate with other devices and/or systems via wireless communications, wired communications, or a combination of wireless and wired communications. For example, communication interface **760** may include mechanisms for communicating with another device or system via a network. Communication interface **760** may include an antenna assembly for transmission and/or reception of RF signals. For example, communication interface **760** may include one or more antennas to transmit and/or receive RF signals over the air. In one implementation, for example, communication interface **760** may communicate with a network and/or devices connected to a network. Alternatively, or additionally, communication interface **760** may be a logical component that includes input and output ports, input and output systems, and/or other input and output components that facilitate the transmission of data to other devices.

(68) Device **700** may perform certain operations in response to processor **720** executing software instructions contained in a computer-readable medium, such as memory **730**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **730** from another computer-readable medium or from another device. The software instructions contained in memory **730** may cause processor **720** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(69) Although FIG. 7 shows exemplary components of device **700**, in other implementations, device **700** may contain fewer components, additional components, different components, or differently arranged components than those depicted in FIG. 7. For example, device **700** may include one or more switch fabrics instead of, or in addition to, bus **710**. Additionally, or alternatively, one or more components of device **700** may perform one or more tasks described as being performed by one or more other components of device **700**.

(70) Systems and methods described herein enable automation of network slice deployments. A device stores a group of preconfigured templates for different slice segments of a data transport network and identifies customer parameters for a slice. The device retrieves a first template that satisfies the customer parameters for a first slice segment and generates a first segment design for the slice. The device retrieves a second template that supports the first segment design and customer parameters for a second slice segment and generates a second segment design for the slice. The device may also retrieve a third template that supports the customer parameters and the second segment design and generates a third segment design for the slice. The device assembles an end-to-end slice design using the first segment design, the second segment design, and the third segment design and forwards the end-to-end slice design to a deployment system for the data transport network.

(71) The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the invention to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of the invention. Also, while a series of blocks have been described with regard to FIG. 6 the order of the blocks and message/operation flows may be modified in other embodiments. Further, non-dependent blocks may be performed in parallel.

(72) Certain features described above may be implemented as “logic” or a “unit” that performs one or more functions. This logic or unit may include hardware, such as one or more processors, microprocessors, application specific integrated circuits, or field programmable gate arrays, software, or a combination of hardware and software.

(73) To the extent the aforementioned embodiments collect, store or employ personal information of individuals, it should be understood that such information shall be collected, stored and used in

accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage and use of such information may be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as may be appropriate for the situation and type of information. Storage and use of personal information may be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

(74) Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another, the temporal order in which acts of a method are performed, the temporal order in which instructions executed by a device are performed, etc., but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

(75) No element, act, or instruction used in the description of the present application should be construed as critical or essential to the invention unless explicitly described as such. Also, as used herein, the article “a” is intended to include one or more items. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

(76) In the preceding specification, various preferred embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Claims

1. A network device comprising: one or more processors configured to: store, in a memory, a plurality of preconfigured templates for different network slice segments of a data transport network; obtain user input to identify customer parameters for a network slice in the data transport network; retrieve, from the plurality of preconfigured templates, a first template that satisfies the customer parameters for a first network slice segment; generate, based on the first template and the customer parameters, a first segment design for the network slice; retrieve, from the plurality of preconfigured templates, a second template that meets the customer parameters for a second network slice segment and supports the first segment design; generate, based on the first segment design, the second template and the customer parameters, at least one second segment design for the network slice; assemble an end-to-end slice design using the first segment design and the at least one second segment design; and forward the end-to-end slice design to a deployment system for the data transport network.
2. The network device of claim 1, wherein the customer parameters include a geographic service location and performance criteria.
3. The network device of claim 1, wherein the one or more processors are further configured to: obtain, after retrieving the first template, additional user input to complete parameters in the first template.
4. The network device of claim 1, wherein the one or more processors are further configured to: retrieve, from the plurality of preconfigured templates, a third template that meets the customer parameters and supports the second segment design; and generate, based on the second segment design, the third template and the customer parameters, a third segment design for the network slice.
5. The network device of claim 4, wherein the first segment design includes a RAN segment design, wherein the at least one second segment design includes a midhaul transport segment design, wherein the third segment design includes a service aggregation point (SAP) transport

segment design with a user plane function (UPF) and session management function (SMF) pair.

6. The network device of claim 1, wherein, when generating the third segment design for the network slice, the one or more processors are further configured to: retrieve a rule set for SAP transport; and instantiate, based on the rule set, a session management function (SMF) and user plane function (UPF) for the network slice.

7. The network device of claim 1, wherein the one or more processors are further configured to: implement the end-to-end slice design in the data transport network.

8. The network device of claim 1, wherein, when generating the at least one second segment design for the network slice, the one or more processors are further configured to: obtain mapping of quality of service (QoS) indicators in a radio access network (RAN) segment to differentiated services code point (DSCP) values in a midhaul transport; and arrange the mappings to a geographic area based on the customer parameters.

9. The network device of claim 1, wherein, when assembling the end-to-end slice design, the one or more processors are further configured to: assign, based on the user input, a type of data traffic for the network slice.

10. The network device of claim 1, wherein, when assembling the end-to-end slice design, the one or more processors are further configured to: obtain, based on the user input, a charging plan for the network slice.

11. The network device of claim 1, wherein, when assembling the end-to-end slice design, the one or more processors are further configured to: retrieve, from the plurality of preconfigured templates, a security template that meets the customer parameters.

12. A method comprising: storing a plurality of preconfigured templates for different network slice segments of a data transport network; obtaining, by a network device of a management network for the data transport network, user input to identify customer parameters for a network slice in the data transport network; retrieving, by the network device and from the plurality of preconfigured templates, a first template that satisfies the customer parameters for a first network slice segment; generating, by the network device and based on the first template and the customer parameters, a first segment design for the network slice; retrieving, by the network device and from the plurality of preconfigured templates, a second template that meets the customer parameters for a second network slice segment and supports the first segment design; generating, by the network device and based on the first segment design, the second template and the customer parameters, at least one second segment design for the network slice; assembling, by the network device, an end-to-end slice design using the first segment design and the at least one second segment design; and forwarding by the network device, the end-to-end slice design to a deployment system for the data transport network.

13. The method of claim 12, wherein obtaining the user input to identify the customer parameters includes: obtaining a geographic service location and a performance criteria for the network slice.

14. The method of claim 12, further comprising: obtaining, after retrieving the first template, additional user input to complete parameters in the first template.

15. The method of claim 12, further comprising: instantiating network functions for the end-to-end slice design in the data transport network.

16. The method of claim 12, wherein generating the second segment design for the network slice further comprises: obtaining mapping of Quality of Service (QoS) indicators in a Radio Access Network (RAN) segment to Differentiated Services Code Point (DSCP) values in a midhaul transport; and arranging the mappings to a geographic area based on the customer parameters.

17. The method of claim 16, wherein arranging the mappings includes: mapping each cell site in the first segment design to a transport aggregation point, wherein the mapping applies Common Language Location Identification (CLLI) codes to identify paths to transport aggregation sites; and mapping each transport aggregation point to a service aggregation point.

18. A non-transitory, computer-readable storage media storing instructions, which, when executed

by one or more processors of a network device, cause the network device to: store a plurality of preconfigured templates for different network slice segments of a data transport network; obtain user input to identify customer parameters for a network slice in the data transport network; retrieve, from the plurality of preconfigured templates, a first template that satisfies the customer parameters for a first network slice segment; generate, based on the first template and the customer parameters, a first segment design for the network slice; retrieve, from the plurality of preconfigured templates, a second template that meets the customer parameters for a second network slice segment and supports the first segment design; generate, based on the first segment design and the customer parameters, at least one second segment design for the network slice; assemble an end-to-end slice design using the first segment design and the at least one second segment design; and forward the end-to-end slice design to a deployment system for the data transport network.

19. The non-transitory, computer-readable storage media of claim 18, wherein the instructions to assemble the end-to-end slice design further include instructions to: assign, based on the user input, a type of data traffic for the network slice; and obtain, based on the user input, a charging plan for the network slice.

20. The non-transitory, computer-readable storage media of claim 18, wherein the instructions to assemble the end-to-end slice design further include instructions to: identify different Operations Support System (OSS) or Business Support System (BSS) parameters for the network slice and match the parameters to an available OSS/BSS profile in the plurality of preconfigured templates.
